21 August 1968

Supersedes
MIL-M-55516(EL)
14 August 1967

MILITARY SPECIFICATION

MICROPHONE, DYNAMIC M-133/U

1. SCOPE

1.1 This specification covers one type of dynamic, moving-coil, noise-cancelling, microphone, designated as above.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on the date of invitation for bids, or request for proposal, form a part of this specification, to the extent specified herein.

SPECIFICATIONS

FEDERAL

L-P-393	Plastic Molding, Polycarbonate Resins, Injection, and Extruded
PPP-B-566	Boxes, Folding, Paperboard
PPP-B-636	Boxes, Fiberboard
PPP-B-676	Boxes, Set-up, Paperboard
PPP-P-291	Paperboard, Wrapping, Cushioning
PPP-T-45	Tape, Gummed-Paper, Reinforced
PPP-T-76	Tape, Pressure-Sensitive, Adhesive Paper
PPP-T-97	Tape, Pressure-Sensitive, Adhesive, Filament-Reinforced

MILITARY

MIL-P-116	Preservation, Methods of
MIL-T-152	Treatment, Moisture and Fungus-Resistant, of Communications, Electronic, and Associated Electrical Equipment
MIL-V-173	Varnish, Moisture and Fungus-Resistant (for the Treatment of Communications, Electronic, and Associated Electrical Equipment)

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MIL-M-13231 Marking of Electronic Items

STANDARDS

MILITARY

MIL-STD-105 Sampling Procedures, and Tables for Inspection of Attributes
MIL-STD-129 Marking for Shipment
MIL-STD-810 Environmental Test Methods

DRAWINGS

Electronics Command

SC-DL-539922 Microphone, Dynamic M-133/U

Ships Systems Command

SK-N-864 Simulated Gun-Blast Producing Equipment

(Copies of specifications, standards, and drawings required by suppliers, in connection with specific procurement functions, should be obtained from the procuring activity, or as directed by the contracting officer.)

2.2 Other publications - The following document forms a part of this specification to the extent herein specified. Unless otherwise indicated, the issue in effect on the date of invitation for bids shall apply.

United States of America Standards Association

Z-24.4-1949 Pressure Calibration of Laboratory Standard Pressure Microphone

(Application for copies should be addressed to the United States of America Standards Association, Incorporated, 10 East 40th Street, New York, New York 10016.)

3. REQUIREMENTS

3.1 Microphone M-133/U is a small high-quality, noise-cancelling, dynamic, unit designed for use within an oxygen mask. It is provided with a short cord and plug (U-173/U). The coil is protected by a moisture barrier seal.

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3.2 Specifications and drawings - Microphone M-133/U shall meet the requirements set forth herein, and shall conform to the design and construction requirements detailed in the drawings listed on SC-DL-539922 for Microphone M-133/U, which constitutes an integral part of the overall specification requirements. Any apparent conflict should be resolved in favor of the specification, per se.

3.2.1 Material - The plastic molding material shall conform to specification L-P-393. The molding process used in the production of the approved preproduction samples shall be completely described, said description being submitted together with the test report of the said samples to the Government. The molding process used and described in said report shall not be changed, or modified, in any manner whatsoever, during any given production run, without specific and written approval of the Government.

3.3 Preproduction samples - The contractor shall supply eighteen preproduction samples for approval, unless otherwise specified. Six samples will be subjected to both Group A and Group B inspection; twelve samples will be subjected to Group C inspection.

3.3.1 Resubmission - Samples subjected to inspection and failing to meet specification requirements may be remade, at the option of the contractor, and resubmitted for inspection. If such sample then conforms to specification requirements, it may be accepted as a unit of production on the contract.

3.3.2 Testing time - The period of testing shall not exceed 60 days.

3.4 Electrical performance -

3.4.1 Impedance - The impedance of the microphone at 20 degrees Centigrade, and 1,000 Hz shall be within the limits of four and seven and one-half ohms.

3.4.2 Dielectric strength - The unit shall withstand the application of 100 volts, DC, when tested as herein stipulated.

3.4.3 Insulation resistance - The unit shall have an insulation resistance of at least one megohm.

3.5 Acoustical performance -

3.5.1 Sensitivity at sea level - The unit shall have a sensitivity of between 2.5 and 5.0 microvolts per dyne, when measured as herein stipulated.

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3 5.2 Sensitivity at 25,000 feet above sea level - At an altitude of 25,000 feet above sea level the sensitivity of the unit shall be within eight decibels of the measured value at sea level

3.5 3 Frequency response - Close sound source - Characteristic at sea level - The frequency-response characteristic of the unit, between 200 and 5,000 Hz, shall be within the limits shown in Figure 1.

3.5 4 Frequency response - Close sound source - Cnaracteristic at 25,000 feet above sea level - The frequency-response characteristic of the unit at an elevation of 25,000 feet above sea level, between 200 and 5,000 Hz, shall be be within the limits shown in Figure 2

3.6 Signal to noise ratio - The signal to noise ratio, defined as the integrated value of the difference in the frequency-response characteristic of the unit to close, and to distant sound, determined as herein stipulated, shall be at least sixteen decibels.

3.7 Linearity - The output of the unit shall be directly proportiona to the input throughout the range of sound pressure levels from 74 to 114 decibels, relative to a sound pressure of 200 microdynes per square centimeter, at any frequency between 200 and 5,000 Hz.

3.8 Harmonic distortion - The harmonic distortion in the output, measured as herein specified, at any frequency between 300 and 3,500 Hz, shall not exceed five percent, with the input sound pressure at a level of 114 decibels, relative to 200 microdynes per square centimeter.

3.9 Endurance - The unit shall be subjected to the endurance test herein specified. The net difference in response after subjection to the said test shall not exceed three decibels, as determined by measurements made both before, and after, said test, nor shall any parts become loose, nor any spurious responses be evident

3.10 Effect of external magnetic field - The effect of an external magnetic field upon the unit in terms of the voltage induced in the voice coil of the microphone shall be determined as herein specified The open circuit voltage shall not exceed 470 microvolts per oersted.

3.11 Effect of stray magnetic field - The magnetic field produced by the microphone itself, when measured as specified herein, shall not cause a deflection of the compass needle in excess of five degrees, at a distance of twelve inches, measured between the nearest edges of the microphone and compass, respectively.

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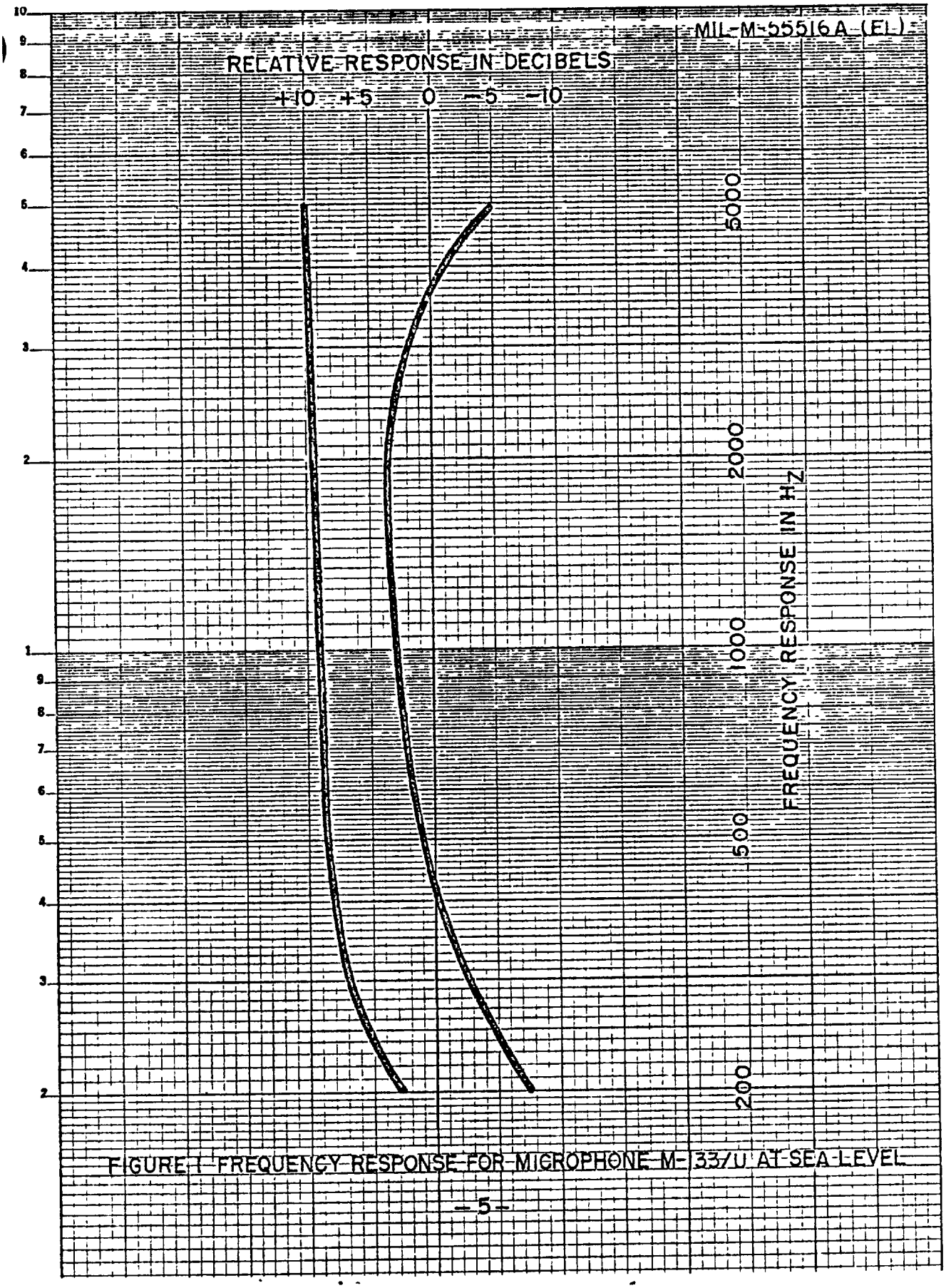


FIGURE 1. FREQUENCY RESPONSE FOR MICROPHONE M-33/U AT SEA LEVEL.

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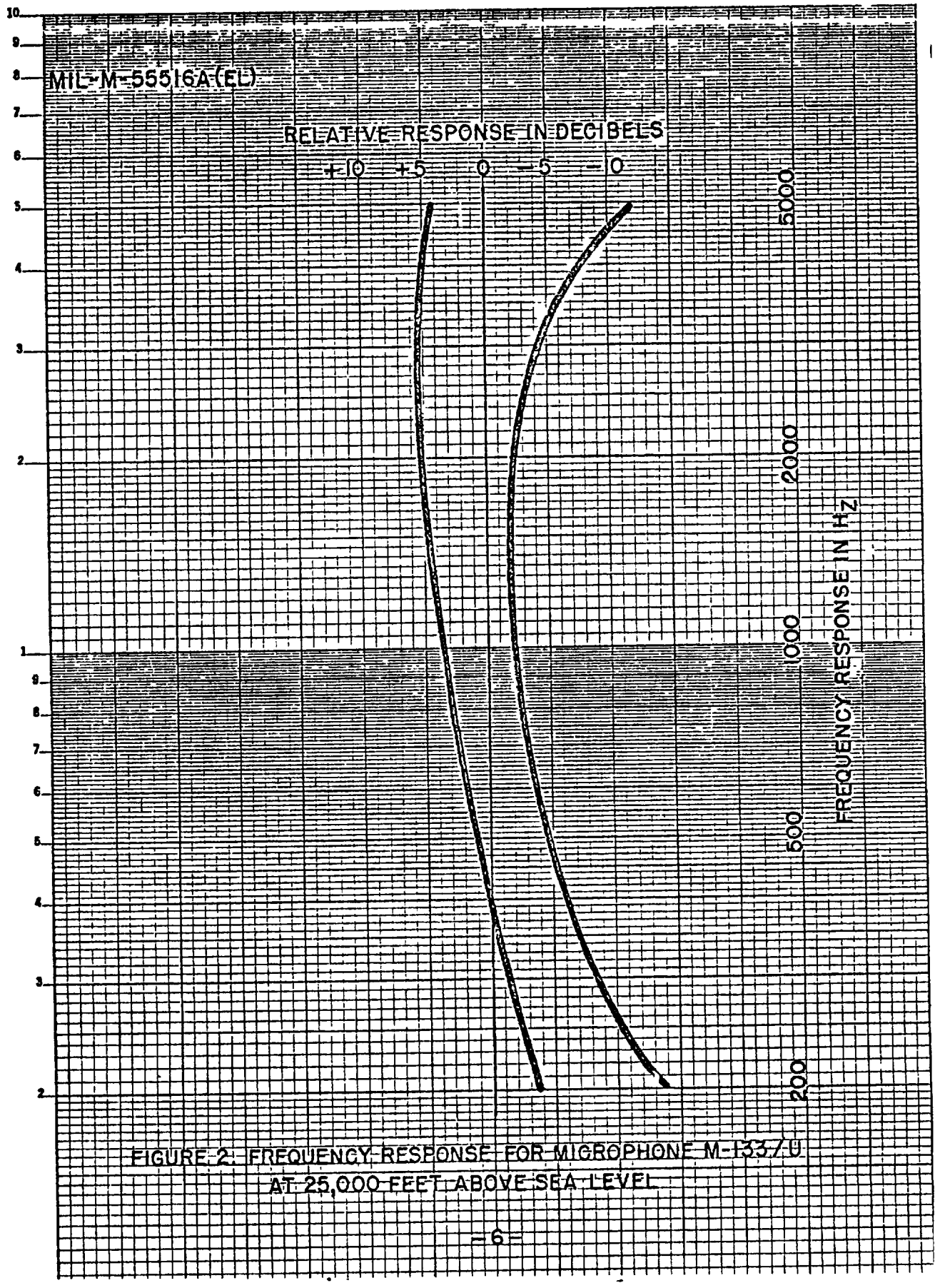


FIGURE 2. FREQUENCY RESPONSE FOR MICROPHONE M-133/U AT 25,000 FEET ABOVE SEA LEVEL

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3.12 Marking - The unit shall be marked in accordance with MIL-M-13231

3.13 Service requirements - The unit shall meet the requirements stipulated herein after subjection to the following environmental tests. Test samples shall be selected as set forth herein. The net change in frequency response shall not exceed three decibels, as determined by measurements made both before, and after each test. The applicable tests are listed in Table III - Group C Inspection.

a. Blast

b. Sand & Dust

c. High Temperature

d. Immersion

e. Humidity

f. Drop

g. Vibration

h. Bounce

i. Elevation

j. Fungus Resistance

3.13.1 Fungus resistance - The unit shall be inert to the attack of fungus. This may be attained by the use of materials which are immune to fungus, or, by subjecting materials, which are not immune to such attack, to appropriate treatment, in accordance with MIL-T-152, and varnish, in accordance with MIL-V-173. In either instance the unit shall be subjected to the test herein stipulated, and meet the applicable criteria.

3.14 Workmanship - The workmanship shall be such as to meet all the requirements of this specification.

4. QUALITY ASSURANCE PROVISION

4.1 Responsibility for Inspection - Unless otherwise specified, the supplier is responsible for all inspection requirements as herein specified. The supplier may utilize his own facilities, or any commercial facility, acceptable to the Government. The Government reserves the right to perform

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any of the inspections stipulated in the specifications, where such inspections are deemed necessary to assure conformance with specified requirements.

4.2 Classification of inspection - Inspection is classified as follows:

a. Preproduction (excluding preparation for delivery).

b. Inspection covered by subsidiary documents.

c. Quality conformance of production.

(1) Equipment inspection, excluding preparation for delivery.

(2) Inspection of preparation for delivery.

4.3 Preproduction inspection - This inspection shall be performed by the contractor; it will be witnessed by a representative of the Government. It consists of all the inspection and test designated as Groups A, B, and C, herein.

4.4 Inspection covered by subsidiary documents - The following shall be inspected under the applicable subsidiary documents:

Item	Reference
Material	3.2.1
Marking	3.12
Fungus Resistance	3.13.1

4.5 Inspection of production units for quality conformance - The contractor shall perform the inspection designated as Group A, B, and C herein. This does not relieve the contractor of his responsibility to perform any additional inspection, which may be necessary to control quality and assure conformance to specification requirements. The Government will examine and evaluate the contractor's inspection procedures and records. In addition, the Government, at its discretion, may perform all, or any part of the inspections specified, in order to verify conformance to specification requirements. Test equipment for such verification shall be made available by the contractor.

4.5.1 Group A inspection - This inspection constitutes a sampling procedure as stipulated in MIL-STD-105. The inspection and tests covered in this group are listed in Table I. (When accomplished as a part of the "preproduction inspection" these measurements will be performed on six sample units).

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TABLE I - Group A Inspection

nspection or Test	Req't Para	Insp Para	AQL% Major	Minor
mpedance	3.4.1	4.8.1	1 dphu*	**
ielectric Strength	3.4.2	4.8.2)	for the	
nsulation Resistance	3.4.3	4.8.3)	group	
isual & Mechanical	3.14	4.7	1 dphu	4 dphu

*Constitutes the group percentage
*All electrical and operating defects are classified as 'Major"

4.5.2 Group B inspection - This inspection also constitutes a sampl- ng procedure as stipulated in MIL-STD-105. The inspection and test covered n this group are listed in Table II. Group B inspection shall be performed n inspection lots which have passed Group A inspection. (When effected as part of the preproduction inspection, these measurements and inspections ill be performed on the same six sample units previously subjected to Group inspection.)

TABLE II - Group B Inspection

nspection or Test	Req't Para	Insp Para	AQL%
ensitivity at Sea Level	3.5.1	4.9.6)	4 dphu for
ensitivity at 25,000 feet above Sea Level	3.5.2	4.9.6)	the group'
ignal to Noise Ratio	3.6	4.10	
inearity	3.7	4.11	
larmonic Distortion	3.8	4.12	
ndurance	3.9	4.13	
ffect of External Magnetic Field	3.10	4.14	
ffect of Stray Magnetic Field	3.11	4.15	

Defects per hundred units

IOTE: The normal inspection level will be S-4.

4.5.3 Group C Inspection - This inspection comprises the tests list- d in Table III, and shall be performed on sample units which have passed oth Group A, and Group B inspections. Sampling comprises two phases, as ollows:

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4.5.3.1 Initial sampling - Twelve samples of the microphone shall be selected at random from the first 1,000 units produced on the order, or contract. These samples comprise the first quality-conformance Group C inspection lot. Eight samples shall be divided into four pairs, and designated as C-1, C-2, C-4, and C-5; the remaining four samples comprise the C-3 group These samples shall be subjected to Group C inspection as shown in Table III

4.5.3.2 Production sampling - Subsequent to the initial sampling, eight samples of the microphone shall be selected at random from each 1,000 units produced on the order, or contract. These samples comprise production, quality-conformance, Group C inspection lots. Four samples shall be divided into two pairs, and designated as C-2, and C-4; the remaining four samples comprise the C-3 group (as before). These samples shall be subjected to Group C inspection as shown in Table III.

4.5.3.3 Orders for less than 1,000 units - Orders for less than 1,000 units shall be subject to sampling and Group C testing in the same manner as is required on orders for more than 1,000 units. The number of units constituting the basis of such inspection, however, is reduced from 1,000 to 100, and a sampling, and Group C inspection is also required on the last 100 units produced on the order.

TABLE III - Group C Inspection

Inspection or Test	Req't Para	Insp Para
C-1		
Blast	3.13	4.16.1
Sand & Dust		4.16.2
C-2		
High Temperature		4.16.3
Immersion		4.16.4
C-3		
Moisture Resistance		4.16.5
Drop		4.16.6
C-4		
Vibration		4.16.7
Bounce		4.16.8
Elevation		4.16.9
C-5		
Fungus Resistance		4.16.10

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4.5.3.4 Group C inspection failure - In the event that a production sample unit fails to pass any of the Group C inspection, or tests, the production lot from which the defective sample was taken shall be set aside. The circumstances attending such failure shall be investigated as soon as possible by the contractor, and the established facts reported to the Government inspector. Pending the determination by the Government inspector, or the contracting officer, as to the significance of such failure, and the remedial action to be taken, acceptance of all units which are suspected of deficiencies, shall be forthwith suspended. Also, any production which is effected subsequent to the date of the Group C inspection failure, will be the contractor's risk; any production lot, or fraction thereof, submitted to the Government for acceptance, will be subjected to whatever inspection procedure may be deemed appropriate by the Government.

4.6 Inspection of preparation for delivery - Preparation for delivery shall conform to the applicable provisions of Specification MIL-P-116, and to the stipulations herein.

4.7 Visual examination of microphone - The microphone shall be visually inspected with particular attention being given to the following factors:

a. Construction and finish of all elements entering into the microphone assembly.

b. Proper assembly of all elements.

c. Cleanliness of all air spaces such as the air gap and vent.

d. Effectiveness of all soldering and sealing operations.

e. Fit and sealing of the microphone case, the moisture barrier to the microphone, and the moisture guard to the moisture barrier.

NOTE: The use of a shadowgraph in determining the cleanliness of the air gap is recommended.

4.8 Electrical performance -

4.8.1 Impedance - The impedance shall be determined by means of any instrument which is capable of effecting this measurement, with no more than ten percent error.

4.8.2 Dielectric strength - The dielectric strength shall be determined by the application of 100 volts DC for a minimum of five seconds, between the voice coil and the metal frame of the microphone element.

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The voice coil shall be energized by the application of 0.005 volt, at 60 Hz thus producing movement of the coil, and the diaphragm elements simultaneously with the application of the test voltage. There shall be no failure of the insulation.

4.8.3 Insulation resistance - The insulation resistance shall be determined by use of an instrument which is capable of measuring one megohm with no more than five percent error. This measurement shall be made after the test for dielectric strength and across the same points.

4.9 Acoustical performance -

4.9.1 Artificial voice - The sound required for the performance of the sensitivity, and frequency-response tests, shall be developed by the Western Electric Company, Nr. 555-W, Artificial Voice, or functional equivalent. This voice shall be calibrated by means of a condenser microphone, such as the(W. E. Co.) 640-AA, which has been previously calibrated in accordance with the reciprocity method specified in A.S.A Standard Z-24.4-1949.

4.9.2 Test fixtures - The fixtures used to position the calibrating microphone, or the test microphone, with respect to the sound source, shall be designed to minimize any possible error in measurements. These may be occasioned by (1) vibration, (2) sound transmitted from the voice tube, (3) unstable positioning of the microphone, (4) reflections from fixtures or environment (such as the interior of an altitude chamber) or (5) ambient noise. Baffles shall not be used near the sound-sensitive parts of the microphone. Test fixtures shall be designed so as to preclude the introduction of errors caused by surface-reflected sound. The microphone under test shall be supported so that its sound port is one-quarter inch from, and coaxial with, the opening in the artificial voice. This distance shall be measured to the microphone case itself, disregarding the moisture barrier, and the moisture-barrier guard.

4.9.3 Sound pressure level - All measurements shall be made at a sound pressure level of 114 decibels relative to 200 microdynes per square centimeter.

4.9.4 Calibration of close sound source - The sound source shall be calibrated to produce a desired sound pressure at the grid of the 640-A-A microphone, over the frequency range of 200 to 5,000 Hz, using the pressure-response calibration of this microphone. The grid of this microphone shall be one-quarter inch from, and coaxial with, the mouth of the artificial voice during calibration. The sound source shall be calibrated at each altitude at which it is used.

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4.9.5 Calibration of distant sound source - The sound field shall be calibrated at a point not less than six feet from the sound source. A Western Electric Company Nr. 640-A-A condenser microphone, or functional equivalent, shall be used employing the free field calibration of the calibrating microphone. The location of the calibrating microphone is known as the calibration point.

4.9.6 Sensitivity - Close Sound Source - The sensitivity of the microphone, at sea level, and at a simulated altitude of 25,000 feet, shall be determined by measurement of the voltage developed by the microphone, across a five ohm resistive load, at a frequency of 1,000 Hz. This voltage shall be measured with a Ballantine Electronic Voltmeter, Model 300, or functional equivalent, having similar ballistic characteristics. The input sound pressure shall be 114 decibels, relative to 200 microdynes per square centimeter.

4.9.7 Frequency Response - Close Sound Source - The frequency-response of the unit shall be determined by measuring the output voltage across a five ohm resistive load, over the frequency range of 200 to 5,000 Hz, at the level specified in 4.9.3. A Ballantine Electronic Voltmeter, Model 300, or functional equivalent, shall be used. Two series of measurements shall be made (1) with the port normally used for talking facing the sound source, and (2) with the rear port facing the sound source. In each case the unit shall be oriented for maximum output. The first series of readings is identified as "Characteristic Response A"; the second series is identified as "Characteristic Response B".

4.9.8 Frequency Response - Distant Sound Source - The frequency-response of the unit with a distant sound source shall be determined in the same manner as with a close sound source, specified in 4.9.7. The readings taken with the port normally used for talking facing the sound source(third series) constitute "Characteristic Response C"; readings taken with the rear port facing the sound source(fourth series) constitute "Characteristic Response D".

4.10 Signal to noise ratio - The "Signal to Noise Ratio" is defined as the index of the ability of the microphone to transmit "signal" i.e. intelligible voice sounds, as distinguished from all other sounds i.e. "noise". This index is determined by an evaluation of the several frequency-response characteristics which have been established as above specified. The evaluation is accomplished graphically in the following manner:

a. The average value of the "Characteristic Response A", and the "Characteristic Response B", determined as specified in 4.9.7 above, shall be computed, and plotted as "Average Close Response E".

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b. The average value of the "Characteriestic Response A", and the "Characteristic Response D", determined as specified in 4.9.8 above, shall likewise be computed, and plotted as "Average Distant Response F".

c. The actual voltages represented by the "Average Close Response E", and the "Average Distant Response F", as measured over that portion of the frequency range in which the signal to noise ratio will be above zero, shall be noted, and the individual voltage ratios for each frequency computed. These voltage ratios shall then be averaged. As the minimum required signal to noise ratio is given as sixteen decibels, this voltage ratio must be at least 6.31 to 1.00, the arithmetical equivalent.

4.10.1 Plotting of the characteristic responses - The response characteristics shall be plotted on semi-logarithmic graph paper, such as Keuffel & Esser 46-4972, or equalivent. (This paper provides for an abscissa of two logarithmic cycles, each five inches wide, and an ordinate of twenty divisions per inch,) Each ordinate division shall represent 0.5 decibel. All points shall be clearly shown, and isolated from each other; the derivation of the signal to noise ratio shall also be plainly indicated.

4.11 Linearity - The output voltage developed by the unit across a five ohm resistive load shall be measured at any one frequency within each of the following three ranges: (1) 200 to 800 Hz, (2) 800 to 2,500 Hz, and 2,500 to 4,000 Hz, produced by a close sound source, with input sound pressures of 74, 94, and 114 decibels, relative to 200 microdynes per square centimeter, respectively.

4.12 Harmonic distortion - The harmonic distortion shall be measured in a manner similar to that employed for the determination of frequency response, except that the Hewlett-Packard Model 330 Distortion Analyzer (Model B, or later), or its functional equivalent, shall replace the voltmeter. The close-source, input sound pressure shall be at a level of 114 decibels, relative to 200 microdynes per square centimeter, at the talking (front port).

4.13 Endurance - An alternating voltage of 0.005 (white noise) in the range from 300 to 3,500 Hz, shall be applied to the unit for three hours, after which the frequency response with a close sound source shall be determined. Compliance with 3.9 is required.

4.14 Effect of external magnetic field - The microphone shall be tested in a uniform magnetic field of ten oersteds, established by a 400 cycle current, as produced by a Helmholtz coil arrangement made by the W. N. Welch Scientific Company, Chicago, Illinois, and identified in Catalog Nr. 623-A. The microphone shall be held in the bare hand in the central volume of the Helmholtz coil arrangement and rotated in all directions in order to determine the maximum induced voltages. These voltages shall be

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expressed in microvolts per oersted. During this test the microphone shall be connected across a 150 ohm load, and care shall be exercised to prevent any contribution to the output voltage by ambient noise. The microphone shall meet the requirements of 3.10.

4.15 Effect of stray magnetic field - This effect shall be determined with the aid of a Keuffel & Esser, Type Nr. 5600 compass, or functional equivalent. The face of the microphone shall be at right angles to the compass needle, and its geometric center approximately twelve inches from the pivot point of the needle. The microphone shall be in the plane of rotation of the needle so that the needle points to the center of the microphone face. The microphone shall be turned about its geometric center in all directions. The maximum needle deflection shall be noted. NOTE: This test shall be made in a location which is substantially free from stray magnetic disturbances. The microphone shall meet the requirements of 3.11.

4.16 Environmental tests - Units, selected as stipulated herein, shall be subjected to the following tests designated as Group C. Measurements of response throughout the designated frequency range shall be made both before and after each test. The net difference in response, in any instance, shall not exceed three decibels, neither shall any external surface or internal surface or component part exhibit any significant deterioration. (Significant deterioration is defined for the purposes of this specification as any visible degradation which can be attributed to flaky, pitted, blistered, or otherwise loosened finish or metal surface).

4.16.1 Blast - The unit shall be mounted on the carriage of the U.S. Navy Simulated Gun-Blast Equipment, in accordance with Ships Systems Command Drawing SK-N-864. The unit shall be secured with the face of the microphone at right angles to the plane of the shock wave, and subjected to 30 blasts, each developing a pressure of 9.5 pounds, peak, per square inch. Measurements of response shall be made both before and after the test. The change in response shall not exceed three decibels.

4.16.2 Sand and dust - The unit shall be subjected to the Dust Test, Method 510, in MIL-STD-810. Reasonable time intervals are permitted between the several steps specified for this test; however, each step shall be performed without any significant interruption. Measurement both before and after the unit is subjected to this test shall be effected.

4.16.3 High temperature - The unit shall be subjected to the High Temperature Test, Method 501, Procedure I in MIL-STD-810.

4.16.4 Immersion - The unit shall be subjected to the Immersion Test, Method 512, Procedure I in MIL-STD-810.

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4.16.5 Humidity - The unit shall be subjected to the first five steps of Procedure III in Method 507 of MIL-STD-810. The unit shall then be subjected to five 48-hour cycles, in immediate succession, as depicted in the figure related to this procedure and shown in the standard. After the cycling the unit shall be subjected to the drop test herein specified.

4.16.6 Drop - The unit shall be dropped twelve times in a random manner from a height of six feet on concrete.

4.16.6.1 Procedure and criteria - The unit shall be tested for response before and after the first five humidity cycles, and first twelve drops. Failure to meet the established criteria, either with respect to the response afforded, or with respect to the physical condition of the unit, constitutes a basis of rejection. The body of the unit shall be substantially intact; only minor chipping and abrasion are tolerable; major fractures, or other significant failure constitutes a basis of rejection.

4.16.7 Vibration - The unit shall be subjected to vibration in accordance with Method 514 of MIL-STD-810, Procedure IX. It shall be firmly secured in a horizontal position and vibrated for the periods specified in the standard.

4.16.8 Bounce - The unit shall be subjected to the following bounce test:

a. Equipment - The equipment used to perform the bounce test shall be the "Package Tester", Type 1, 000-SC, as made by the L.A.B. Corporation, situated at Skaneateles, New York, or functional equivalent.

b. Surface preparation - Cover the planking with one-half inch plywood so that the grain of the wood is parallel to the chain drive. Fasten the plywood with sixpenny nails spaced about six inches apart, in a six inch square, in the center. Use a nailset to drive the nails well below the surface.

c. Compartment - Construct one open compartment, using one-half inch plywood for the handset. The horizontal dimensions should provide for lateral motion of from one to three inches of the handset.

d. Operation - Operate the machine, shafts in phase, for three hours, at 284 ±2 rpm.

4.16.9 Elevation - The unit shall be subjected to the Altitude Test, Method 500, Procedure I in MIL-STD-810.

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4.16.10 Fungus resistance - The unit shall be subjected to a 28 day Fungus-Resistance Test, in accordance with Method 508 of MIL-STD-810.

5. PREPARATION FOR DELIVERY

5.1 Level A (See 6.2)

5.1.1 Cleaning - The unit shall be cleaned as stipulated in Process C-1 of MIL-P-116.

5.1.2 Drying - The unit shall be dried as stipulated in MIL-P-116.

5.1.3 Unit packaging - Units shall be packaged Method IC-1. The microphone cable shall be wound around the unit itself. The unit shall be cushioned by wrapping in paperboard conforming to Type I of Specification PPP-P-291. The wrapping shall be secured by tape, Type III, per Specification PPP-T-45. After units have been sealed inside of a IC-1 bag they shall be placed within a close-fitting box conforming to Variety 1 of PPP-B-566, or to Grade A per PPP-B-676. The box shall be closed in accordance with the provisions in the appendix of the applicable box specification.

5.1.4 Intermediate packaging - Unit packages, prepared as above stipulated, shall be placed in intermediate containers conforming to Variety 2 of Specification PPP-B-566. Such containers shall be uniform in size and shape, shall have minimum cube and tare, shall contain multiples of five units, and shall not exceed ten pounds each. Such containers shall be closed as provided in the appendix to the applicable box specification. All seams and joints shall be sealed with tape, not less than two inches wide, conforming to PPP-T-76.

5.2 Level C-Packaging - Each unit shall be preserved and packaged in a manner that will afford adequate protection against corrosion, deterioration, and damage during shipment, from the supply source to the first receiving activity (See 6.2).

5.3 Level A - Packing - Units, packaged as stipulated in 5.1 or 5.2 (Level A or C), shall be packed in a close-fitting, fiberboard box, conforming to PPP-B-636, Type CF, weather-resistant class, for special requirements use. Closure shall be effected in accordance with the provisions in the appendix of the applicable box specification. The shipping container shall be reinforced by the application of pressure-sensitive, filament tape, conforming to PPP-T-97, Type IV, as specified in the appendix of the applicable box specification (See 6.2).

5.4 Level B - Packing - Units, packaged as stipulated in 5.1 or 5.2 (Level A or C), shall be packed as specified in 5.3, except that the fiber

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board box shall be domestic class, and the strapping is not required. (See 6.2)

5.5 Level C - Packing - Units, packaged as stipulated in 5.1 or 5.2 Level A or C), shall be packed in shipping containers, in a manner that will afford adequate protection against damage, either to the package or to its contents, during shipment from the supply source to the first receiving activity. Shipping containers shall conform to the rules and regulations of the common carrier, as applicable to the mode of transportation employed (See 6.2).

5.6 Marking - Marking shall conform to MIL-STD-129, and to the stipulations (if any) in the contract.

6. NOTES

6.1 Intended use - Microphone M-133/U is intended for use in Army Aircraft Oxygen Masks.

6.2 Procurement data - Procurement data should include the following:

a. Number, title, and date of this specification.

b. Levels of packaging and packing required.

c. Whether the packaging and packing must withstand the rough handling tests of MIL-P-116.

d. Marking and shipping of preproduction samples.

e. The place designated for final inspection.

6.3 Nomenclature - After the approval of the preproduction samples, the contractor should apply to the contracting officer for the assignment of the approved nomenclature.

6.4 Corrosion criteria - The corrosion criteria herein defined, as applicable, are supplemental to the criteria defined in MIL-STD-810.

6.5 Test conditions and test equipment - All test conditions specified herein, and in subsidiary documents, must be strictly adhered to, within the specified tolerances. Violations of specified test requirements may constitute grounds for the rejection of units subjected to test conditions which exceed the tolerances specified, in any respect, and in any test.

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All proposed substitutions of test equipment, or modifications of test procedure must be approved by the Government.

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Project Nr. 5965-A083

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No 22-R255

INSTRUCTIONS This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost Comments and the return of this form will be appreciated Fold on lines on reverse side, staple in corner and send to preparing activity Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements

SPECIFICATION

MIL-M-55516A(EL) MICROPHONE, DYNAMIC M-133/U

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes" in what way?)*

4 REMARKS *(Attach any pertinent data which may be of use in improving this specification If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED

ESC-FM 1068-68

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U.S.Army Electronics Command
Fort Monmouth, New Jersey 07703

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